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Course: BTech Cse Data Science

Semester: 5th

Subject: Statistical Foundation of Data Science

Explanation Of Code :

First, I imported Pandas, NumPy and Matplotlib. Then I loaded my Excel file into a DataFrame. After that, I checked the dataset using head, shape, columns and info. Then I calculated descriptive statistics like mean, median, minimum and maximum. I created a histogram for the beauty variable and compared beauty scores between male and female professors. Finally, I used cross-tabulation to find the percentage of tenured professors by gender.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_excel('/Professor_Rating_Small_Dataset.xlsx')
```

```
pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)

df
```

	Professor_ID	gender	tenure	students	beauty
0	1	Male	Yes	35	0.45
1	2	Female	Yes	28	0.72
2	3	Male	No	42	-0.15
3	4	Female	Yes	25	0.60
4	5	Male	Yes	50	0.10
5	6	Female	No	31	0.85
6	7	Male	Yes	45	-0.30
7	8	Female	No	22	0.55
8	9	Male	No	38	0.05
9	10	Female	Yes	29	0.78
10	11	Male	Yes	55	-0.20
11	12	Female	Yes	34	0.68
12	13	Male	No	41	0.25
13	14	Female	Yes	27	0.90
14	15	Male	Yes	48	-0.05

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.head()
```

	Professor_ID	gender	tenure	students	beauty
0	1	Male	Yes	35	0.45
1	2	Female	Yes	28	0.72
2	3	Male	No	42	-0.15
3	4	Female	Yes	25	0.60
4	5	Male	Yes	50	0.10

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

```
(15, 5)
```

```
df.columns
```

```
Index(['Professor_ID', 'gender', 'tenure', 'students', 'beauty'], dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  -
```

```
0 Professor_ID 15 non-null int64
1 gender       15 non-null object
2 tenure       15 non-null object
3 students     15 non-null int64
4 beauty       15 non-null float64
dtypes: float64(1), int64(2), object(2)
memory usage: 732.0+ bytes
```

```
df.describe()
```

	Professor_ID	students	beauty
count	15.000000	15.000000	15.000000
mean	8.000000	36.666667	0.348887
std	4.472136	9.933110	0.413450
min	1.000000	22.000000	-0.300000
25%	4.500000	28.500000	0.000000
50%	8.000000	35.000000	0.450000
75%	11.500000	43.500000	0.700000
max	15.000000	55.000000	0.900000

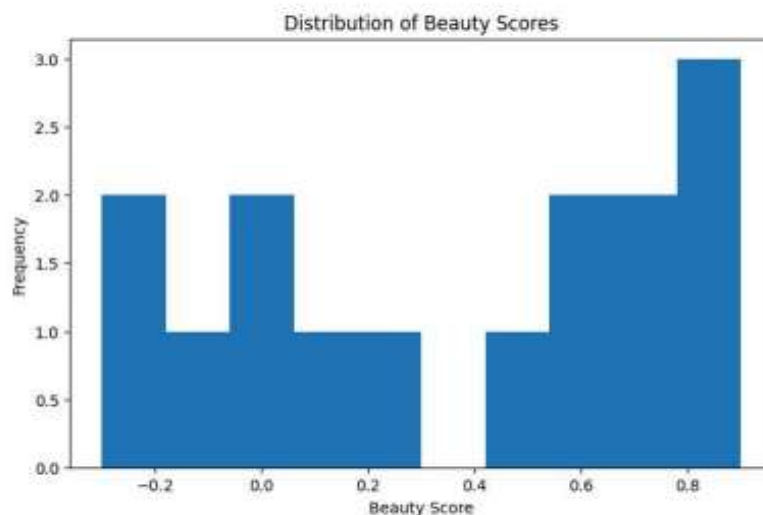
```
print("Mean:", df['students'].mean())
print("Median:", df['students'].median())
print("Minimum:", df['students'].min())
print("Maximum:", df['students'].max())
```

```
Mean: 36.666666666666664
Median: 35.0
Minimum: 22
Maximum: 55
```

```
plt.figure(figsize=(8,5))
plt.hist(df['beauty'].dropna(), bins=10)

plt.xlabel('Beauty Score')
plt.ylabel('Frequency')
plt.title('Distribution of Beauty Scores')

plt.show()
```



```
df['gender'].unique()
```

```
array(['Male', 'Female'], dtype=object)
```

```
beauty_gender = df.groupby('gender')['beauty'].agg(['mean', 'std'])  
beauty_gender
```

	mean	std
gender		
Female	0.725714	0.127522
Male	0.018750	0.247758

Next steps: [Generate code with beauty_gender](#) [New interactive sheet](#)

```
df['tenure'].unique()
```

```
array(['Yes', 'No'], dtype=object)
```

```
pd.crosstab(df['gender'], df['tenure'])
```

	tenure	No	Yes
gender			
Female		2	5
Male		3	5

```
tenure_percentage = pd.crosstab(  
    df['gender'],  
    df['tenure'],  
    normalize='index'  
) * 100
```

```
tenure_percentage
```

	tenure	No	Yes
gender			
Female		28.571429	71.428571
Male		37.500000	62.500000

Next steps: [Generate code with tenure_percentage](#) [New interactive sheet](#)